

An L^1 Obstruction to Naive Stone–von Neumann Uniqueness for Weyl Pairs

arXiv automatic math run `fa_banach_001`

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Status

This packet gives a candidate partial obstruction to Open Problem (4) of van Neerven–Portal [1]. It does not prove a Banach-space Stone–von Neumann theorem. It shows that the most literal Hilbert-style analogue, where irreducible Weyl pairs are expected to be similar to Hilbert standard-pair models, is impossible in the full class of Banach spaces.

Source Question

At the end of the paper the authors list four open problems:

9. OPEN PROBLEMS

As explained in the introduction, this paper is mostly meant as a foundation for the development of pseudo-differential calculi in “rough” settings. We nonetheless

think that the general theory of Weyl pairs presented here is also worth developing further in its own right. This would include solving the following problems:

- (1) To extend Mauceri’s results on twisted convolutions [39] to Bochner spaces $L^p(\mathbb{R}^{2d}; X)$, where X is UMD and $p \in (1, \infty)$.

An affirmative answer would automatically solve the next problem.

- (2) To extend Theorem 4.9 to general Weyl pairs (A, B) .

As observed in Remark 8.4, for standard pairs on $L^p(\mathbb{R}^d; X)$ with $1 < p < \infty$, the conclusions of Theorems 7.1, 8.1, and 8.5 hold for arbitrary UMD Banach spaces X . In the three theorems for general Weyl pairs, the lattice structure of X was only used through the proof of Lemma 7.2. Thus one may pose the following problem:

- (3) To decide whether Theorems 7.1, 8.1, and 8.5 hold for arbitrary UMD spaces X .

An affirmative answer to the first problem could possibly also solve the third problem, since it might pave the way for an alternative proof via transference. For these three problems, studying the particular case where X is a non-commutative L^p -space would be particularly interesting, yet potentially much simpler than the general case (thanks to the availability of both domination and extrapolation techniques).

- (4) To prove an analogue of the Stone–von Neumann uniqueness theorem for Weyl pairs.

The target here is item (4): prove an analogue of the Stone–von Neumann uniqueness theorem for Weyl pairs.

Definitions Used

For the standard position/momentum pair on $L^1(\mathbb{R}^d)$, write

$$(M_u f)(x) = e^{iu \cdot x} f(x), \quad (T_v f)(x) = f(x + v), \quad u, v, x \in \mathbb{R}^d.$$

These are the bounded C_0 -groups $e^{iu \cdot Q}$ and $e^{iv \cdot P}$ from Example 3.2 of the source. A closed subspace $Y \subseteq L^1(\mathbb{R}^d)$ is called invariant for the standard Weyl pair if $M_u Y \subseteq Y$ and $T_v Y \subseteq Y$ for every $u, v \in \mathbb{R}^d$. The pair is topologically irreducible if the only closed invariant subspaces are $\{0\}$ and $L^1(\mathbb{R}^d)$.

Idea of the Proof

The invariant-subspace part is a time-frequency cyclicity argument. If a nonzero $f \in L^1$ is fixed, then the closed span of all modulates and translates $M_u T_v f$ is all of L^1 . Indeed, any annihilator

$h \in L^\infty$ would make the Fourier transform of $x \mapsto f(x+v)h(x)$ vanish for every translation v , so $f(x+v)h(x) = 0$ almost everywhere for every v . A Fubini intersection argument then forces $h = 0$. This proves irreducibility.

The obstruction to Hilbert-style uniqueness is then purely Banach-geometric: $L^1(\mathbb{R}^d)$ is non-reflexive, while any Hilbert direct sum of standard L^2 -models is reflexive.

Lemma 1 (Cyclicity of one nonzero vector). *Let $0 \neq f \in L^1(\mathbb{R}^d)$. Then*

$$\overline{\text{span}}\{M_u T_v f : u, v \in \mathbb{R}^d\} = L^1(\mathbb{R}^d).$$

Proof. Let Z denote the closed span on the left. Suppose $h \in L^\infty(\mathbb{R}^d)$ annihilates Z , in the usual duality

$$\langle g, h \rangle = \int_{\mathbb{R}^d} g(x)h(x) dx.$$

Then for all $u, v \in \mathbb{R}^d$,

$$0 = \langle M_u T_v f, h \rangle = \int_{\mathbb{R}^d} e^{iu \cdot x} f(x+v)h(x) dx.$$

For each fixed v , the function $x \mapsto f(x+v)h(x)$ belongs to $L^1(\mathbb{R}^d)$. By uniqueness of the Fourier transform on $L^1(\mathbb{R}^d)$, it follows that

$$f(x+v)h(x) = 0$$

for almost every x , for every $v \in \mathbb{R}^d$.

Assume, toward a contradiction, that $h \neq 0$. Choose $\alpha, \beta > 0$ and a ball $B_R \subset \mathbb{R}^d$ such that the sets

$$E = \{y : |f(y)| > \alpha\}, \quad F = \{x \in B_R : |h(x)| > \beta\}$$

have positive finite measure. This is possible because $f \neq 0$ in L^1 and $h \neq 0$ in L^∞ . By Fubini's theorem,

$$\int_{\mathbb{R}^d} |F \cap (E - v)| dv = |F| |E| > 0.$$

Hence for some v , the set $F \cap (E - v)$ has positive measure. On this set $|h(x)| > \beta$ and $|f(x+v)| > \alpha$, contradicting $f(x+v)h(x) = 0$ almost everywhere. Thus $h = 0$.

The annihilator of Z in L^∞ is zero. By Hahn–Banach, $Z = L^1$. $\square$

Theorem 1. *For every $d \geq 1$, the standard position/momentum Weyl pair on $L^1(\mathbb{R}^d)$ is topologically irreducible.*

Proof. Let $Y \subseteq L^1(\mathbb{R}^d)$ be a nonzero closed subspace invariant under all M_u and T_v . Choose $0 \neq f \in Y$. Then $M_u T_v f \in Y$ for all $u, v \in \mathbb{R}^d$, and by the lemma their closed linear span is all of $L^1(\mathbb{R}^d)$. Hence $Y = L^1(\mathbb{R}^d)$. $\square$

Corollary 1 (Obstruction to a naive Hilbert-style analogue). *There is an irreducible Banach-space Weyl pair which is not similar to a direct sum of Hilbert-space standard position/momentum pairs.*

Proof. Take the standard pair on $L^1(\mathbb{R}^d)$. By Example 3.2 of the source it is a Weyl pair, and by the theorem it is topologically irreducible.

The Banach space $L^1(\mathbb{R}^d)$ is non-reflexive. For instance, it contains a closed isometric copy of ℓ^1 , obtained from functions supported on pairwise disjoint measurable sets of finite positive measure. Since ℓ^1 is non-reflexive, $L^1(\mathbb{R}^d)$ is non-reflexive as well. Any Hilbert direct sum of standard L^2 -models is a Hilbert space, hence reflexive. Therefore there is no Banach-space isomorphism from $L^1(\mathbb{R}^d)$ onto such a Hilbert direct sum, and consequently no similarity intertwining the corresponding Weyl pairs. $\square$

Limitations

This is only a partial obstruction. It does not rule out a Stone–von Neumann analogue formulated inside a restricted Banach category, such as Hilbert spaces, certain L^p -spaces with fixed p , reflexive UMD spaces, Banach lattices with extra structure, or non-commutative L^p -spaces. It also does not address the vector-valued twisted-convolution problems (1)–(3).

Novelty Check

The local indexes were searched for arXiv:1806.00980, Weyl calculus, Weyl pairs, group generators, canonical commutation relations, Stone–von Neumann, twisted convolution, UMD, and functional-calculus keywords. No exact prior packet for this open-problem item was present.

External searches on June 28, 2026 for combinations of the source title, Stone–von Neumann, Weyl pairs, Banach spaces, twisted convolution, and UMD did not locate a later solution to the source open problems. Since this packet is a partial obstruction rather than a full solution, the novelty claim is modest: it records a simple boundary condition that any positive analogue must respect.

Human Review Recommendation

Verify that topological irreducibility is an acceptable Banach-space analogue of Hilbert irreducibility for the intended discussion. If the intended open problem already restricts to Hilbertizable or reflexive settings, this packet should be read only as a scope warning. Under the literal wording, it is a valid obstruction to a naive Hilbert-model uniqueness theorem.

References

- [1] J. van Neerven and P. Portal, *The Weyl calculus for group generators satisfying the canonical commutation relations*, arXiv:1806.00980.